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EDUCATION

Doctor of Philosophy, Electrical and Computer Engineering <i>University of Windsor, ON</i>	Sept 2023 – Present
Master of Applied Science, Electrical and Computer Engineering <i>University of Windsor, ON</i>	May 2019 – Dec 2021
Bachelor of Engineering, Electronics and Communication Engineering <i>Anna University, India</i>	Aug 2014 – Apr 2018

WORK EXPERIENCE

Doctoral Teaching and Research Assistant, BMS Lab <i>University of Windsor, ON</i>	Sept 2023 – Present
◦ Ph.D. research centered on Li-ion battery modeling, state estimation, and control strategies to enhance the safety, performance, and reliability of energy storage systems.	
Full-time Research Assistant, BMS Lab <i>University of Windsor, ON</i>	Aug 2022 – Aug 2023
◦ Led battery data planning, collection, and analysis in MATLAB to advance battery SOx estimation algorithms. ◦ Worked with a multidisciplinary research team to prototype a wireless battery management system.	
Master's Teaching and Research Assistant, Mechatronics Lab <i>University of Windsor, ON</i>	May 2019 – Dec 2021
◦ Master's research centered on design and development of novel feature-based occupancy map-merging algorithm to improve autonomous multi-robot navigation.	

RECOGNITION & SCHOLARSHIPS

Entrance Scholarship (CDN \$7500) <i>University of Windsor, Canada</i>	May 2019
Best Student Paper Award <i>International Conference on Innovative Research in Engineering and Science, Thailand</i>	Jun 2017

PROFESSIONAL MEMBERSHIPS & CERTIFICATIONS

Student Member

Sept 2023 – Present

Institute of Electrical and Electronics Engineers (IEEE)

Lithium Battery Safety

Oct 2023

Lion Technology Inc.

MENTORING

Graduate Mentor, University of Windsor Formula Electric Team

Jan 2023 – Present

University of Windsor, ON

- A humble part of an incredible journey — from battery module design (2023) to a fully functional EV battery pack (2025).
- Mentored a multidisciplinary undergraduate engineering team in the design and testing of a high-performance EV battery pack and battery management system.

Graduate Mentor, University of Windsor Capstone Teams

Jan 2023 – Present

University of Windsor, ON

- FSAE Accumulator Design (2025) — Ali Elshikh, Zuhair Abdo, Sahana Jayatilaka, Jared Brouerius Van Nidek
- SOH Indicator for a Battery Fuel Gauge (2024) — Sam Letwin, Nicolas Manzon, Andy Garro
- Power System Modeling for the Formula Electric Team (2023) — Saad Arshad, Shaan Batra, Aleks Dejanovic, Luka Tojcic
- Ionic Wind Pump for Battery Cooling (2023) — Sabreen Elastal, Asra Kala, Simrat Poonia, Matthew Aguilar, Boron Al-Diraie
- Active Cell Balancing for an EV Battery Pack (2023) — Justin Haddad, Luca Mannarino, Nick Nguyen

SYNERGISTIC ACTIVITIES

Conference Judge, UWill Discover Student Research Conference

Mar 2026

University of Windsor, ON

Peer Reviewer (9 Reviews for 4 Journals)

2023 – Present

Journal of Energy Storage (4), Journal of Power Sources (3), Measurement (1), MethodsX (1)

PUBLICATIONS

Journals:

- [J6] Sunil, S., Pattipati, K.R., and Balasingam, B. (Under Review). Multiple-Model Adaptive Equivalent-Circuit Identification for Li-ion Batteries via Two-Stage Least Squares. *IEEE Transactions on Industrial Electronics*.
- [J5] Sunil, S., Sundaresan, S., Pattipati, K.R., and Balasingam, B. (Under Review). The Role of Open-Circuit Voltage Characterization in Model-Based Capacity Estimation. *IEEE Transactions on Instrumentation and Measurement*.
- [J4] Sunil, S., Pillai, P., Pattipati, K.R., and Balasingam, B. (2025). Two-Stage Least Squares for Equivalent-Circuit Parameter Estimation of Lithium-Ion Batteries Using Pulse-Relaxation Excitation. *IEEE Transactions on Instrumentation and Measurement*.
- [J3] Pillai, P., Desai, S., Sunil, S., Pattipati, K.R., and Balasingam, B. (2025). Least Squares Approach to Improve Standardized Estimation of OCV and Internal Resistance in Batteries. *IEEE Journal of Emerging and Selected Topics in Industrial Electronics*.
- [J2] Sunil, S., Pattipati, K.R., and Balasingam, B. (2025). Piecewise Linear Approximation of Battery Open-Circuit Voltage Characteristics Using Dynamic Programming. *IEEE Transactions on Instrumentation and Measurement*.
- [J1] Sunil, S., Mozaffari, S., Singh, R., Shahrrava, B., and Alirezaee, S. (2023). Feature-Based Occupancy Map-Merging for Collaborative SLAM. *Sensors*, 23(6), 3114.

Conferences:

- [C11] Sunil, S., and Balasingam, B. (Accepted). Deep Learning vs. Model-Assisted Learning: Objective Performance Evaluation of the LSTM Classifier Using Time Series Data. In *2026 29th International Conference on Information Fusion*.
- [C10] Sunil, S., Pattipati, K.R., and Balasingam, B. Improved Linear Estimation of Battery Open-Circuit Voltage and RC Model Parameters from Pulse-Relaxation Data. In *2025 IEEE 4th Industrial Electronics Society Annual On-Line Conference (ONCON)* (pp. 1–6). IEEE.
- [C9] Nguyen, N., Sunil, S., Pillai, P., and Balasingam, B. An Offsetting-Based Correction to Improve the Accuracy of Low-Rate OCV Curves for Lithium-Ion Batteries. In *2025 IEEE 34th International Symposium on Industrial Electronics (ISIE)* (pp. 1–6). IEEE.
- [C8] Mahmoodzadeh, M., Sunil, S., Pillai, P., Biondi, F., Strayer, D.L., Cooper, J.M., McDonnell, A.S., and Balasingam, B. Temporal Analysis of Cognitive Workload During Manual and Partial Driving Automation Based on the Detection Response Task. In *2025 IEEE 34th International Symposium on Industrial Electronics (ISIE)* (pp. 1–5). IEEE.
- [C7] Sundaresan, S., Sunil, S., Balasingam, B., and Pattipati, K.R. Joint Estimation of Open-Circuit Voltage and Equivalent Circuit Model Parameters Using State-Space Model Optimization. In *2023 IEEE Transportation Electrification Conference & Expo (ITEC)* (pp. 1–6). IEEE.
- [C6] Sunil, S., Sundaresan, S., Pillai, P., and Balasingam, B. Inverse Characterization of Open-Circuit Voltage for State-of-Charge Estimation of Batteries. In *2023 IEEE Transportation Electrification Conference & Expo (ITEC)* (pp. 1–6). IEEE.
- [C5] Sunil, S., Balasingam, B., and Pattipati, K.R. State-of-Charge Estimation of Batteries Using the Extended Kalman Filter: Insights into Performance Analysis and Filter Tuning. In *2022 IEEE 1st Industrial Electronics Society Annual On-Line Conference (ONCON)* (pp. 1–6). IEEE.
- [C4] Sunil, S., Mozaffari, S., Singh, R., Shahrrava, B., and Alirezaee, S. LiDAR-Based Cooperative SLAM with Different Parameters. In *2022 7th International Conference on Mechanical Engineering and Robotics Research (ICMERR)* (pp. 82–87). IEEE.
- [C3] Sundaresan, S., Sunil, S., Balasingam, B., and Pattipati, K.R. Fast Offline Battery Capacity Estimation Approach with Performance Bounds. In *2022 IEEE Electrical Power and Energy Conference (EPEC)* (pp. 188–193). IEEE.
- [C2] Sunil, S., Sundaresan, S., Balasingam, B., and Pattipati, K.R. Novel Table-Based Kalman Filter for State-of-Charge Estimation of Batteries. In *2022 IEEE Electrical Power and Energy Conference (EPEC)* (pp. 200–205). IEEE.
- [C1] R Dhaya, T. R. Nirmal, M. Sudharsun, S. Sooraj, D. Kanthavel (Best Paper Award). Resourceful and Proficient Exploitation of MDYMO in Mobile Adhoc Networks In *2017 International Conference on Innovative Research in Engineering and Science (IRES)*

Theses:

- [T2] S. Sunil. “Heterogeneous Collaborative Mapping for Autonomous Mobile Systems”, *University of Windsor, Canada*, Jan. 2022.
- [T1] S.R. Abhirami, S. Sooraj, R. Subashini, K. Venkatesh.. “LISI: Language Interpreter for Speech Impaired”, *Anna University, India*, Apr. 2018.